Historical Returns

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Returns

- Return R_{t+1} at time $t+1$ is defined

$$R_{t+1} = \frac{P_{t+1} - P_t}{P_t}$$

- where P_t Price at time t
- Example:
  - Buy stock at $P_t = \$100$
  - Sell stock at $P_{t+1} = \$110$
  - Return is $R_{t+1} = \frac{P_{t+1} - P_t}{P_t} = \frac{\$110 - \$100}{\$100} = 10\%$

Decompose Returns

$$R = R_f + RP$$

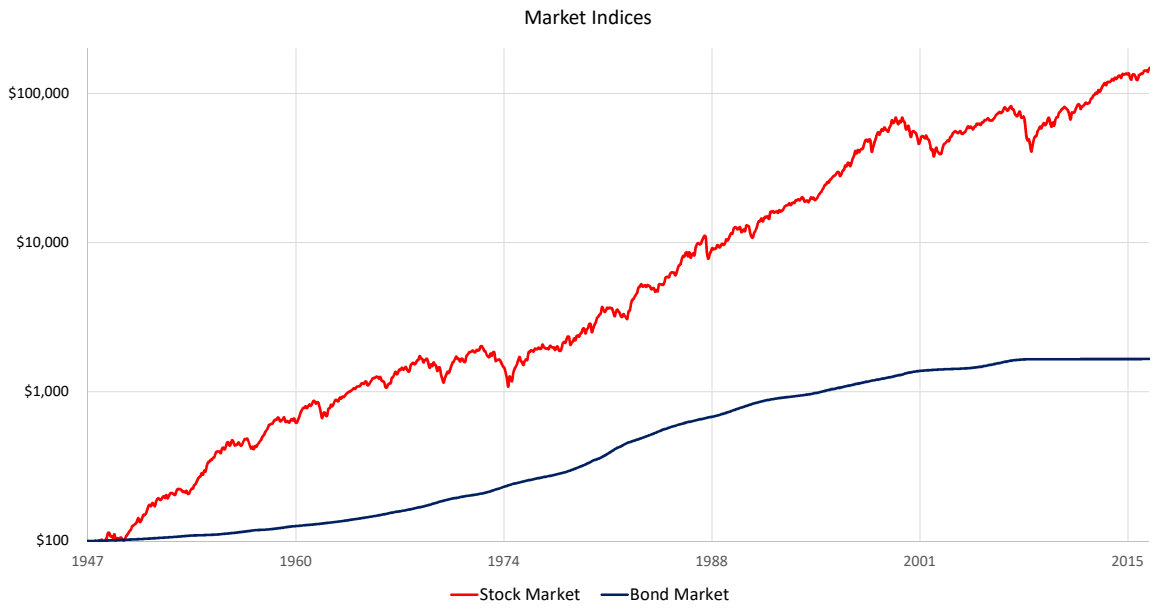
- Risk-free rate R_f
  - The risk-free rate reflects the time value of money
  - How willing an investor is to give up $1 today for $$$ in the future
  - The more impatient a consumer, the more future compensation he will demand to delay consumption and hence the higher the risk-free rate
- Risk premium RP
  - Investors who dislike risk are risk averse and demand risk premiums for bearing risk
  - The risk-premium increases when:
    - The riskiness of an asset increases
    - Investors become more risk averse

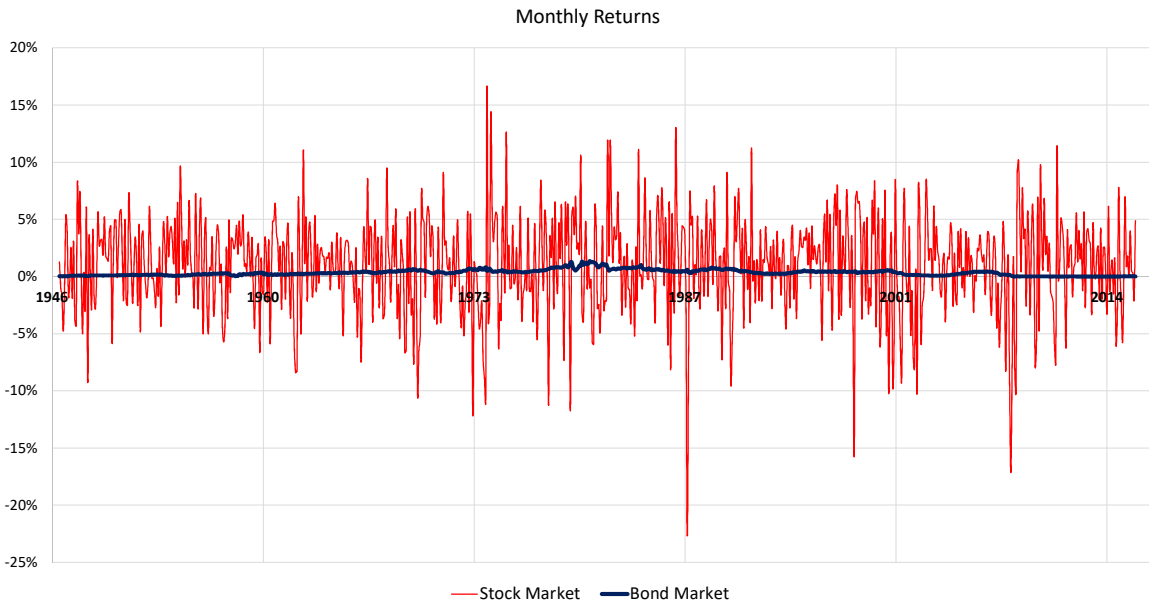
Empirical Risk Premiums

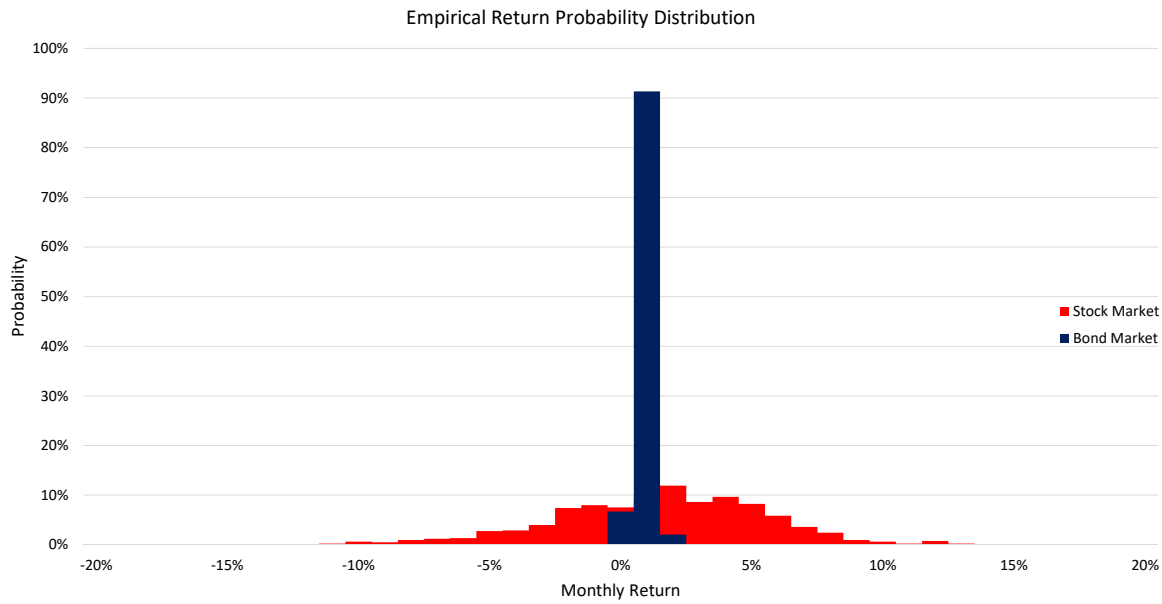
- Measure risk premiums in the data by looking at the difference between an asset's expected return and the risk-free rate

$$RP = E[R] - R^f$$

- We seek to understand what are the important risk factors that determine risk premiums
- Bad stocks have high returns
  - High returns reflect compensation for time and risk


Market Indices
$100,000
$10,000
$1,000
$100
1947
1960
1974
1988
2001
2015
Stock Market
Bond Market


Monthly Returns
20%
15%
10%
5%
0%
-5%
-10%
-15%
-20%
-25%
1946
1960
1973
1987
2001
2014
Stock Market
Bond Market



Sharpe Ratio: Reward-to-Risk

- Excess Return: $EXR_t = R_t - R_f$
  - The difference between an asset's return and the risk-free rate at time t
- Risk Premium: $RP = E[EXR_t] = E[R_t] - R_f$
  - The expected (i.e. average) excess return
- Sharpe Ratio
  - The ratio of an asset's risk premium to standard deviation
  - Represents reward-risk tradeoff

$$Sharpe\ Ratio = \frac{E[R_t] - R_f}{\sigma} = \frac{RP}{\sigma}$$

Note on Trading Strategy

- The excess return $EXR_t = R_t - R_f$ represents the return on a trading strategy where
  - Buy (long) stock
  - Sell (short) risk-free bond
    - Selling a bond is the same as borrowing at the risk-free rate

- Today:
  - Sell 3% bond for $100 (i.e. borrow $100 at 3% interest)
  - Buy $100 of stock
- 1 year later:
  - Repay bond loan $103
  - Assume stock went up 10% to $110
  - Sell stock for $110
  - Net: $110-$103=$7
  - $7 represents 10%-3%=7% excess return of stock over bond

Normal Distribution & Finance

- Common to assume returns are normally distributed
  - Mean is a measure of reward
  - Standard deviation is a measure of risk
  - No theoretical justification for assumption
  - Empirically, Normality is close and is a good starting point for financial models
  - However, in the data we do observe
    - Negative skewness
    - Positive kurtosis
  - This implies we underestimate tail risk in our financial models
  - To the extent returns are not normally distributed
    - Standard deviation is no longer a complete measure of risk
    - Sharpe ratio is not a complete measure of portfolio performance

Risk Analysis & Value at Risk (VaR)

- p VaR – Loss such that there is a probability p of observing at least as extreme

- If asset has 5% VaR of $1 million, there is a 5% probability of the asset losing $1 million or more

Implications

Stat	Finance	Observe	Implication
Mean	Reward	7.5%	Double wealth ~10 yr
Standard Deviation	Risk	15%	95% of years have returns [-22%,37%]
Skewness	Disaster Tail Risk	Negative	Underestimate Risk
Kurtosis	Fat Tail Risk	Positive	Underestimate Risk
Sharpe	Reward-Risk Ratio	0.50	Benchmark to grade other investments